07/09/26

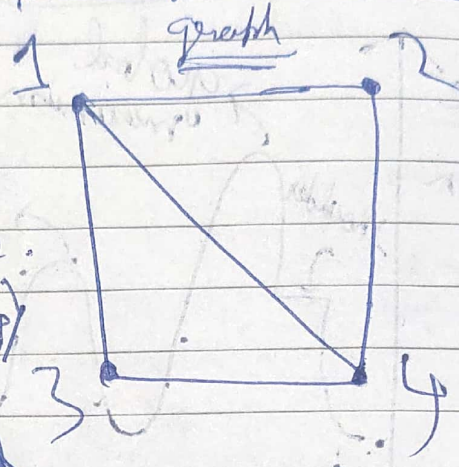
Constraint Satisfaction Problem

Domain = {R, G, B}

V = {1, 2, 3, 4}

no adjacent vertice of same color.

C = {1≠2, 1≠3, 3≠4, 2≠4, 1≠4}



constraints

straight cut way

Initial domain	1	2	3	4
	RGB	RGB	RGB	RGB
1 = R	R	GB	GB	GB
2 = G	R	G	GB	B
3 = B	R	G	B	~~B~~
3 = G	R	G	G	B

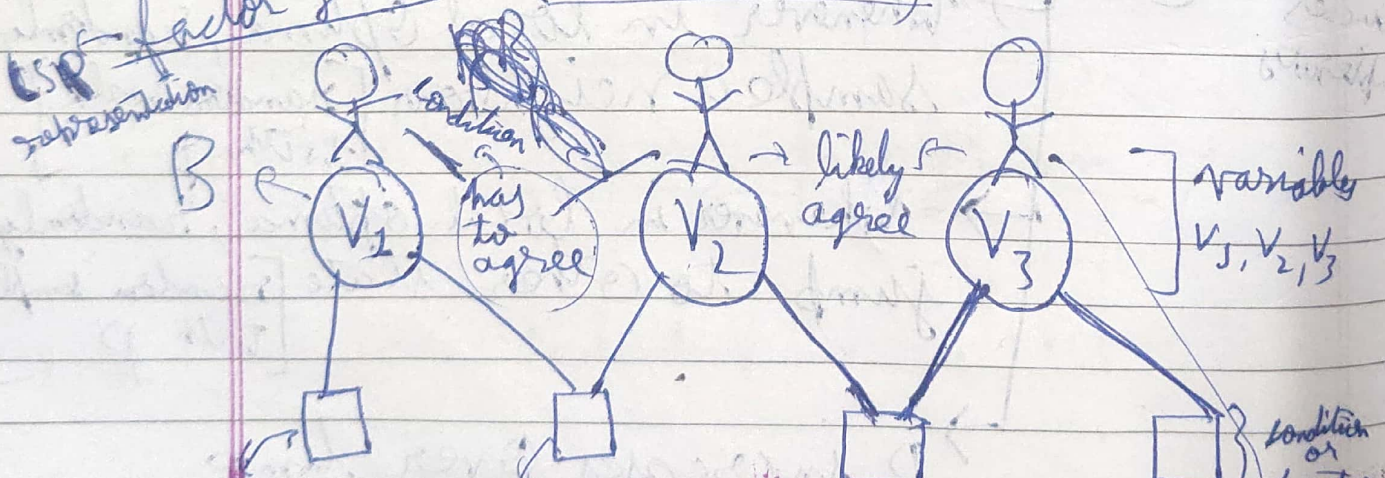
Not possible backtrack

~~Different problem~~

factor graph

assume only B, R

CSP representation



x_1	$f_1(x_1)$
R	0
B	1

$f(x_1) = [x_1 = B]$

x_1	x_2	$f_2(x_1, x_2)$
R	R	1
R	B	0
B	R	0
B	B	1

x_2	x_3	$f_3(x_2, x_3)$
R	R	2
R	B	1
B	R	1
B	B	2

fixed

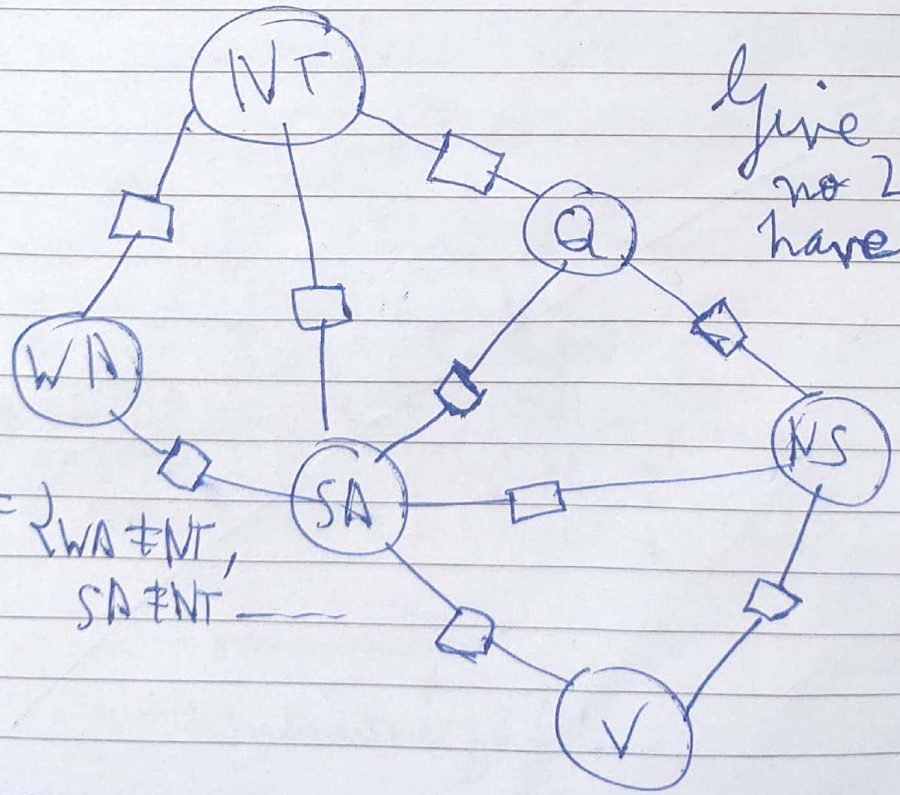
x_3	$f_3(x_3)$
R	2
B	1

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we have defined in previous factor graph

x_1	x_2	x_3	Weight → multiply, then if weight > 0 it is a soln
R	R	R	0, 1, 2, 2 = 0
R	R	B	= 0
R	B	R	= 0
R	B	B	= 0
B	R	R	1, 0, 2, 2 = 0
B	R	B	1, 0, 1, 1 = 0
B	B	R	1, 1, 1, 2 = 2
B	B	B	1, 1, 2, 1 = 2

Q- Eg



Give such that no 2 adj. vertices have same color.

Constraint = {WA ≠ NT, SA ≠ NT ---

List

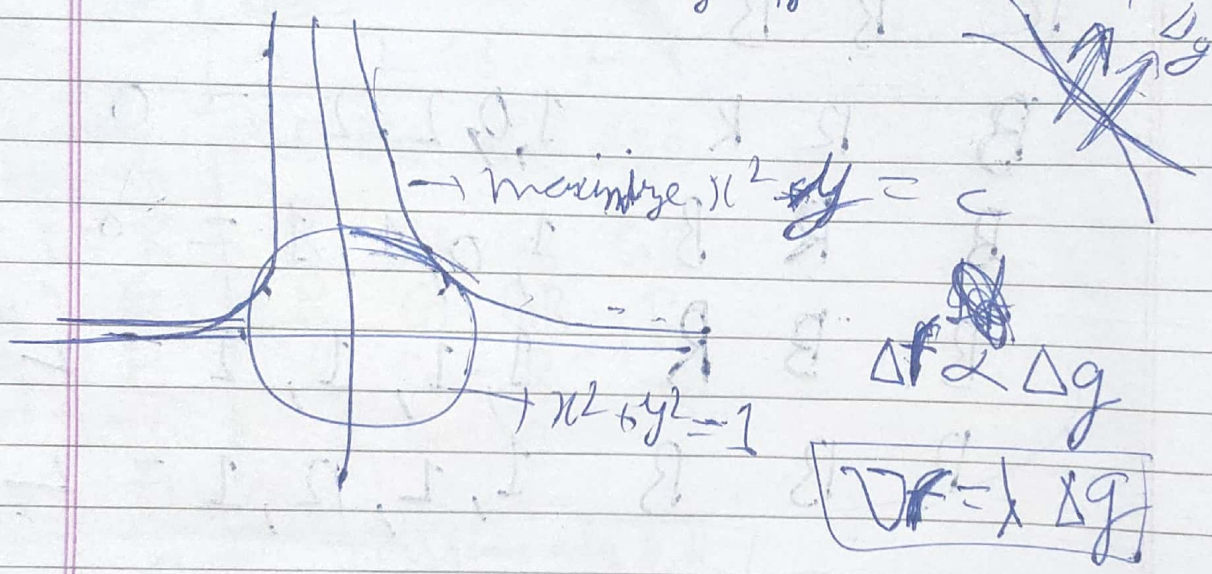
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Constraint optimization problem

Eg - Maximize $f(x,y) = x^2 y$

On the set $x^2 + y^2 = 1$

$g(x,y)$



$\Delta f \propto \Delta g$

$\boxed{\nabla f = \lambda \Delta g}$

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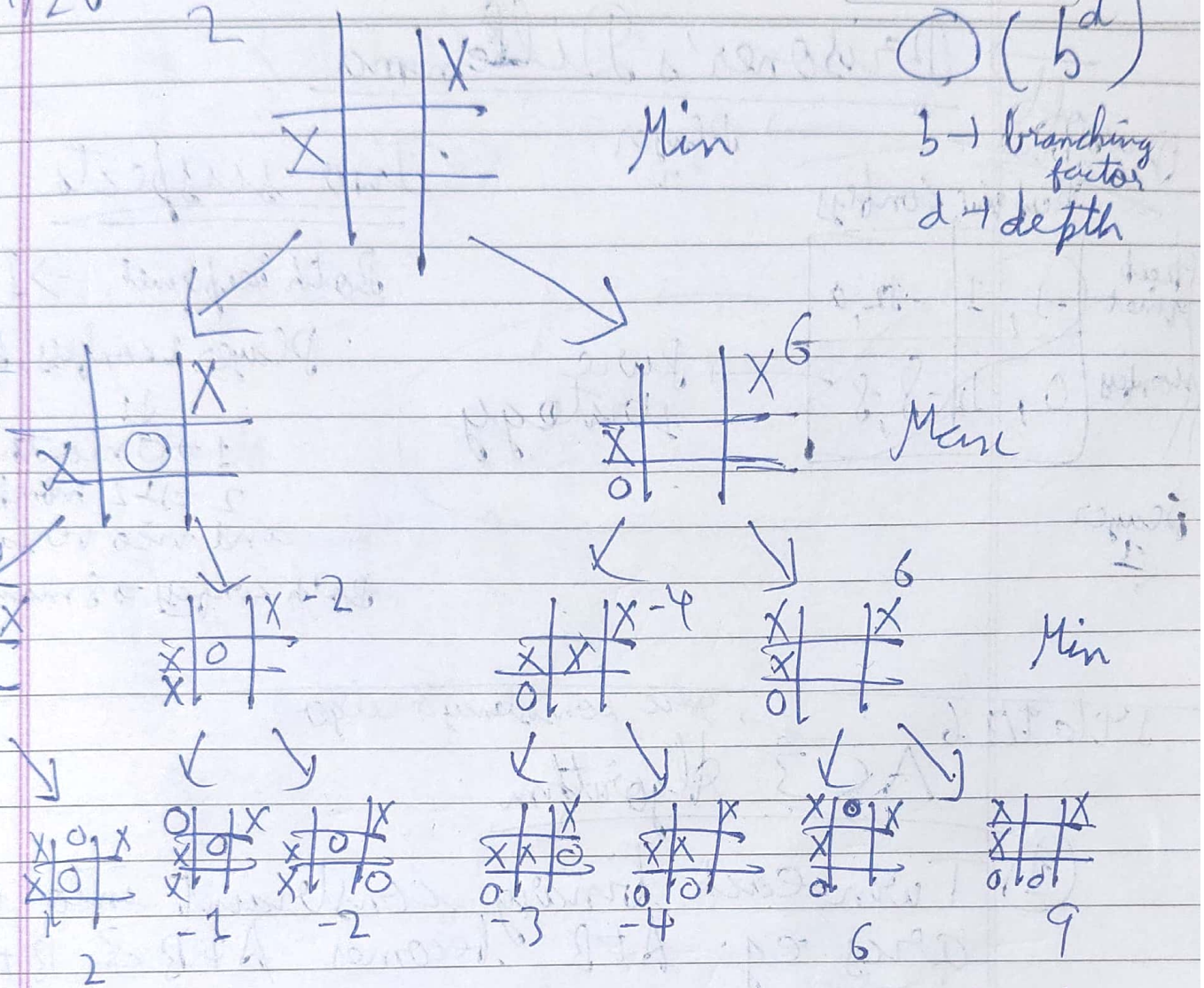
Min - Max algorithm

09/09/26

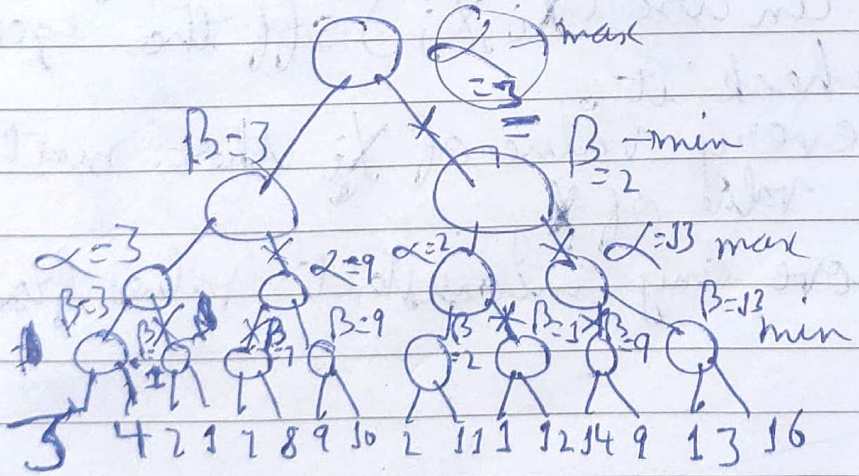
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$O(b^d)$

b → branching factor
d → depth



To reduce T.C. → α-β pruning



if α ≥ β
↳ stop exploring

α → maximizes
β → minimizes

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Prisoner's Dilemma

Payoff matrix

Player 1 \ Player 2	keep quiet	confess
keep quiet	-1, -1	-12, 0
confess	0, -12	-8, -8

→ Pure strategy

Two suspects

- Both keep quiet → 1 month
- Player 1 confess, Player 2 keep quiet
  ↓
  1 ⇒ 0 month
  2 ⇒ 12 months
  and vice versa
- Both confess ⇒ 8 months.

14/04/26 → Arc consistency 3 algo

AC3 Algorithm

1. Turn each binary constraint into two arcs eg. $A \neq B$ becomes $A \neq B$ & $B \neq A$
2. ⇒ Add all the arcs to an agenda
   ⇒ take an arc (X_i, X_j) off the agenda and check it.
   ⇒ for every value of X_i there must be some value of X_j
   ⇒ remove any inconsistent values from X_i.

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Eg- A = {1, 2, 3}
B = {1, 2, 3}
C = {1, 2, 3}

A > B
B = C

Agenda	Arcs
A > B (remove 1)	A > B
B < A (remove 3)	B < A
B = C	B = C
C = B (remove 3)	C = B
A > B	
B < A	
B = C	
C = B	

Eg- X = {1, 2}
Y = {1, 2}
empty Z = {1, 2}
X ≤ Y
Y ≤ Z
Z < X

Agenda	Arcs
X ≤ Y (remove 2)	X < Y
~~X ≤ Y~~	Y > X
Y ≥ X	X < Z
Y < Z X	Z > Y
~~Z > Y~~	Z < X
~~Z ≤ X~~	X > Z
X ≥ Z	

Backtracking method
X = 1 X < Y
X = 1 fails
backtracks
Y = 2 → valid
Z = 1 → fails
Z = 2 → fails
X = 2 ⊗ Backtracks

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Player 1 \ Player 2	go	stop
go	-5, -5	1, 0
stop	0, 1	-1, -1

→ can stick to single strategy

→ Nash Equilibrium

↓

Red light

→ A Nash equilibrium is a set of strategies, one for each player, such that no player has incentive to change his/her strategy independent of others.

Zero sum game

	Heads	Tails
Heads	1, -1	-1, 1
Tails	-1, 1	1, -1

→ can't stick to single strategy

$\frac{1}{2}$ of time $\frac{1}{2}$ of time

16/09/26

Propo

→ A pro statement false.

Eg: The

→ A comp of pro connecti (∨, ∧,

→ Tautolog matter proposi

→ The are ca denoted

p ≡

orga

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Propositional Logic

→ A proposition is a declarative statement that is either true or false.

Eg: The sky is blue.

→ A compound proposition is composed of propositions and one or more connectives like

$(\vee, \wedge, \rightarrow, \leftrightarrow)$ T F

→ Tautology: A compound proposition that is always true, no matter what the truth values of propositional variables

→ The compound proposition p & q are called "logically equivalent" when denoted by '≡' notion

$p \equiv q$

→ Demorgan's law: $\sim(p \wedge q) \equiv \sim p \vee \sim q$

$\sim(p \vee q) \equiv \sim p \wedge \sim q$

Eg: "Miguel has a cell phone (p) & he has a computer (q)"

$p \wedge q$

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

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Contradiction → always false

Contingency → neither true nor false.

Eg:- $p \to q$ is contingency.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

☆ Satisfiability

* A compound proposition is satisfiable if there is an assignment of truth values to its variables that makes it true.

$\Sigma = (P \vee Q) \wedge (\neg P)$

p = False, q = true → true — satisfiable

Predicate logic

x is less than 10

Q(x)

Q(x=3) → True

Q(x=11) → false

$Q(x) \wedge P(x)$

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• n-ary predicate :

A statement of the form $P(x_1, \ldots, x_n)$ is the value of the propositional function P at the n-tuple $(x_1, x_2, \ldots, x_n)$ and called an n-ary predicate.

$Q(x, y)$ = true if $x < 0$ & $y > 5$

↳ 2-ary predicate.

• Argument

→ An argument is collection of declarative statement where one statement is claimed to follow the others.

→ An argument form replaces concrete propositions with propositional variables & logical connectors ($\wedge, \vee, \sim, \rightarrow, \leftrightarrow$).

☆ Modus Ponens

If p then q } → Premises → If they are true

p

∴ q → Conclusion → then this is true

Argument form consist of two premises & a conclusion.

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2
given below premises tell what can be concluded.

Q- ① If the sum of the digits of 371,487 is divisible by 3, then 371,487 is divisible by 3 → q
② The sum of digits of 371,487 is divisible by 3 → p

Ans. p → q : true
p : true

then q is true, i.e. 371,487 is divisible by 3 is true.

☆ Modus Tollen

If p then q } premises → if true
~ q
~ p → conclusion → then it is true

Eg: ① If 870,232 is divisible by 6, then it is divisible by 3. (p → q)
② 870,232 is not divisible by 3 (~ q)
∴ conclusion → 870,232 is not divisible by 6. (~ p)

$\forall x \in D$ if $P(x) \to Q(x)$

contrapositive $\Rightarrow \sim Q(x) \to \sim P(x)$

$Q(x) \to P(x)$

$\sim P(x) \to \sim Q(x)$

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Quantifier

Universal ($\forall$)

Existential ($\exists$)

$\forall x \in D$ if $P(x) \to Q(x)$

contrapositive : $\sim Q(x) \to \sim P(x)$

converse : $Q(x) \to P(x)$

Inverse : $\sim P(x) \to \sim Q(x)$

Eg: $\forall x \in \mathbb{R}$ if $x > 2$ then $x^2 > 4$

contrapositive : $\forall x \in \mathbb{R}$ if $x^2 \leq 4$, then $x \leq 2$.

Note: $p \to q \equiv \sim p \vee q$

18/09/16
Gate 2014

$S_1 : \sim p \wedge (p \vee q) \to q$

$S_2 : q \to (\sim p \wedge (p \vee q))$

Check if S_1 & S_2 are tautology or not.

Sol: S_1 & S_2

$\Rightarrow \sim p \wedge (p \vee q)$

$\Rightarrow (\sim p \wedge p) \vee (\sim p \wedge q)$

$\Rightarrow F \vee (\sim p \wedge q)$

$\Rightarrow \sim p \wedge q$

$S_1 : \sim p \wedge q \to q$; $S_2 : q \to \sim p \wedge q$

$S_1 \Rightarrow \sim(\sim p \wedge q) \vee q$

$\Rightarrow (p \vee \sim q) \vee q$

$\Rightarrow p \vee T \Rightarrow T$

$S_2 \Rightarrow \sim q \vee (\sim p \wedge q)$

$\Rightarrow (\sim q \vee \sim p) \wedge (\sim q \vee q)$ $\to T$

$\Rightarrow \sim q \vee \sim p$

Gate 2006

Q - "Some real no's are not rational numbers".

$R(x)$ = "x is a real no."

$Q(x)$ = "x is a rational no."

Make it using $\exists$, $\forall$

Ans. "There exists at least one x, such that x is a real no. and x is not a rational no."

~~$\exists x : R(x) \wedge \sim Q(x)$~~

$\exists x : R(x) \wedge \sim Q(x)$

Q - Check satisfiability → make P true

$P = (A \vee \sim B) \wedge (B \vee C) \wedge (\sim A \vee \sim C)$

T T T

$A \to T, B \to T, C \to F$

Gate 2013

Q - "Gold and Silver are precious metal"

$G(x)$: "x is gold"

$S(x)$: "x is silver"

$P(x)$: "x is precious metal"

Ans. $\forall x \; G(x) \vee S(x) \to P(x)$

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Gate 2013

Q - "Not every student who takes the test passes it".

$S(x)$: "x is student"
$T(x)$: "x takes the test"
$P(x)$: "x passes the test".

Ans. $\sim(\forall x\ S(x) \wedge T(x) \rightarrow P(x))$

if entire expression is tautology

* Valid argument : $P_1, P_2, \ldots, P_n \subset$

$P_1, P_2, P_3, \ldots P_n \rightarrow C$

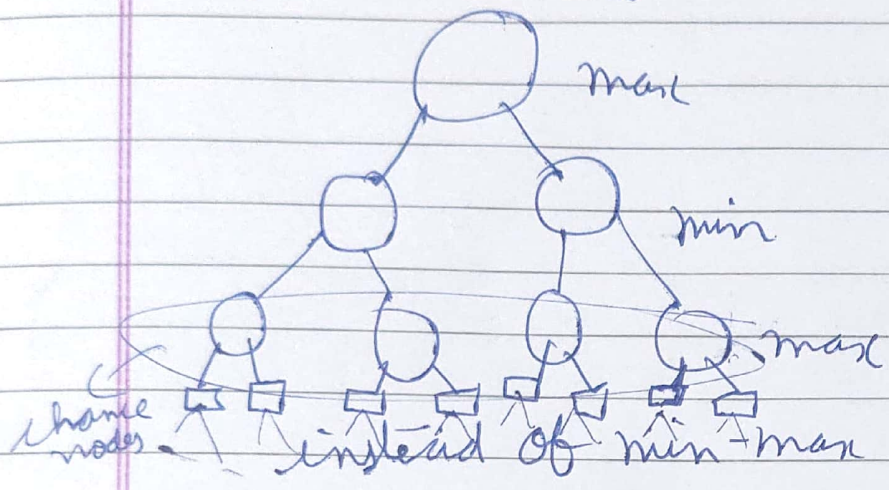
Q = $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

(is tautology (check))

Ans $((\sim p \vee q) \wedge (\sim q \vee r)) \rightarrow (p \rightarrow r)$

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* Expectimax → chance node (extra layer)
↳ avg. of nodes.
↳ then minmax



instead of min-max ⇒ avg.

19/09/26